FRONT-END DEVELOPMENT
HACKATHON – Level-2 Template (Interactivity Implementation)

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TITLE: ACTIVITY TRACKER PRO

GITHUB LINK: https://github.com/kishorebtsa/ACTIVITY-TRACKER-PRO.git

1. Objective

The objective of this project is to implement front-end interactivity that allows users to update the status of activities and view real-time progress. This is achieved using JavaScript or React by handling user actions, updating the UI dynamically, and reflecting changes instantly without requiring a page reload.

2. Problem Focus

In Level-1, the application consisted of a static user interface where data was displayed but not interactive. The challenge in Level-2 is to transform this static UI into a dynamic system by introducing:

- Event handling (user actions like button clicks)
- State changes (updating activity status)
- Dynamic UI updates (real-time rendering)

This transition is important to create a responsive and user-friendly experience.

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3. Application Overview

The application is an **Activity Tracker** that allows users to:

- Add new activities (participants)
- View them in a list/table
- Mark activities as completed
- Track progress in real-time

This helps simulate real-world task management systems.

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4. Data Handling Approach

Data is managed using:

- **React state (useState)** to store activities
- **Local Storage** to persist data even after page reload
- Each activity is stored as an object with properties like:
  - name
  - email
  - phone

- event
- mode
- completed (boolean)

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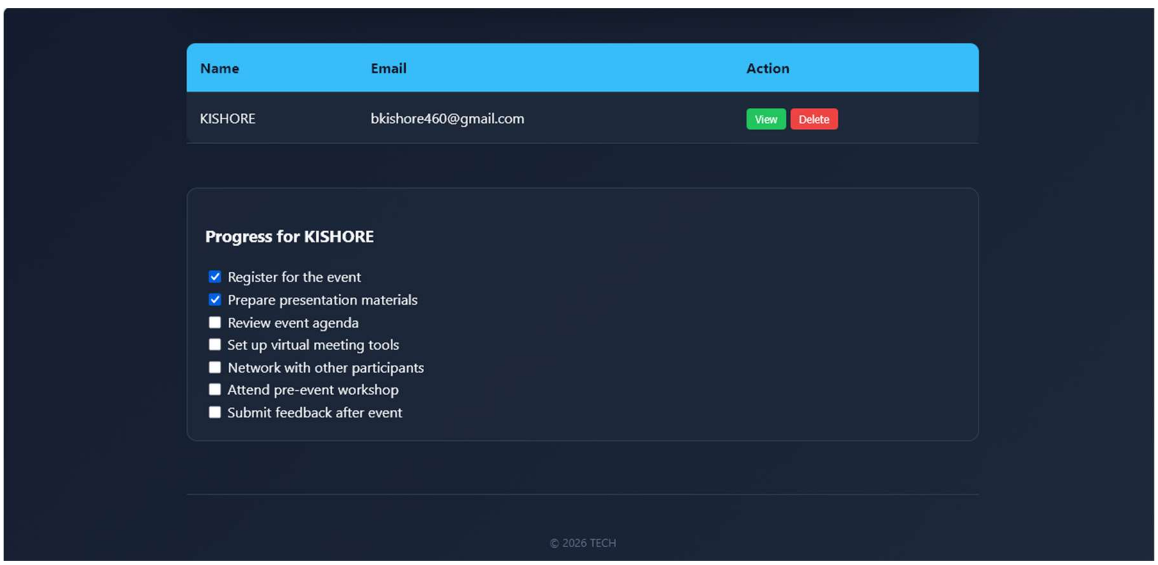
5. UI Components Design

The application is divided into key components:

- **Form Component** → To add new activities
- **Activity List / Table Component** → Displays all activities
- **Buttons / Controls** → Mark as completed, delete
- **Progress Component** → Displays completion status (e.g., "3 out of 5 completed")



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6. Interactivity Implementation

Interactivity is implemented using:

- Event listeners (onClick, onSubmit)
- Updating activity status from **Pending → Completed**
- Triggering re-render of UI using React state updates

Example:

- Clicking “Mark Complete” updates the activity status
- UI updates instantly without reload

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7. Application Logic

Core logic includes:

- Counting completed activities using filter:

participants.filter(p => p.completed).length

- Updating progress dynamically:
    - “X out of Y completed”
- Automatically re-rendering UI when state changes

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8. Technology Used

- **HTML** → Structure
- **CSS** → Styling
- **React JS** → Interactivity and state management
- **JavaScript (ES6)** → Logic and event handling

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9. Best Practices Followed

- Used meaningful variable and function names
- Avoided hardcoding values
- Separated components for clean structure
- Added reusable logic
- Handled edge cases:
    - No activities

    - All activities completed

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10. Key Features Implemented

- Mark activity as completed
- Delete activity
- Real-time progress tracking
- Dynamic UI updates without reload
- Data persistence using Local Storage

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11. Expected Output / Deliverables

- Fully working front-end application
- Interactive activity tracking system
- Real-time progress display
- Clean and structured source code
- GitHub repository with README

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12. Learning Outcome

Through this project, the following concepts were learned:

- Event handling in JavaScript/React
- DOM manipulation / React rendering
- State management using useState
- Building interactive UI components
- Using Local Storage for persistence

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13. Conclusion

This project successfully transforms a static interface into an interactive application. By implementing event handling, state updates, and dynamic rendering, the application provides a seamless user experience. It demonstrates the importance of interactivity in modern web applications and builds a strong foundation for advanced front-end development.